AgentProof

The Trust Oracle for the ERC-8004 Agent Economy

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agentproof.sh

Abstract

Scalar reputation systems are failing the agent economy. AgentProof replaces static scores with adaptive, probabilistic trust -- treating every agent as a probability distribution, not a number. The oracle indexes 54,000+ agents across 21 chains, evaluates trust through an 11-signal composite scoring model with Bayesian smoothing, detects risk through 14 automated flags, and provides dollar-denominated max exposure ceilings for insurance underwriting. This paper documents the architecture, methodology, evidence base, and commercial model.

21 Chains | 54K+ Agents | 11 Scoring Signals | 14 Risk Flags | 6 Pricing Tiers

1. The Crisis of Static Trust

As autonomous agents become economic participants -- executing trades, processing claims, negotiating terms -- they inherit reputation systems designed for human e-commerce. Star ratings. Thumbs up. Transaction counters. These systems are failing.

Current reputation models in ERC-8004 rely on Scalar Accumulation: simple counters that measure volume, not certainty. While this works for slow-moving human markets, it is catastrophically insufficient for high-speed agent economies where thousands of transactions occur per hour and a compromised agent can drain liquidity in minutes.

There is a deeper problem: agents cannot spot a fake review. Humans have intuition, social context, and the ability to read between the lines of a suspicious 5-star rating. Agents have milliseconds and numbers. They will consume whatever score they are given and act on it at machine speed. If the reputation layer is wrong, the damage propagates through the network before any human can intervene. This makes the oracle not just useful but existential infrastructure -- what DNS is to the internet, trust scoring is to the agent economy.

1.1 Scalar Blindness

A new agent with 5 successful transactions often looks identical to a veteran with 5,000. Accumulative systems measure volume, not certainty. It is mathematically impossible to distinguish between 'High Potential' and 'High Reliability', leading to misallocated capital and misplaced trust.

1.2 The Exit Scam Problem

Static scores are sticky. If an agent spends months building a perfect reputation and then turns malicious, a scalar system is too slow to react. Lifetime averages hide recent behaviour. A compromised agent can drain liquidity for days before their score drops.

1.3 The Sybil Vulnerability

In a permissionless system, creating a new identity is nearly free. Bad actors can spin up thousands of bot wallets to wash-trade and artificially inflate scores. Most systems cannot distinguish between 100 reviews from 100 unique users and 100 reviews from a single bot farm.

1.4 Binary Thinking in a Probabilistic World

Current systems ask: 'Is this agent good?' This is the wrong question. An agent might be 99% reliable at token transfers but only 60% reliable at complex arbitrage. A single static score cannot capture this multidimensional reality. Trust decisions require probability distributions, not binary labels.

1.5 Agent-Specific Behavioural Patterns

Research on LLM-driven agents shows they behave fundamentally differently to humans in trust scenarios. GPT-4 based agents are 'unforgiving' -- a single bad interaction can permanently alter their cooperation strategy. Reputation systems must model these agent-specific behavioural clusters, not assume human-like forgiveness curves.

> *"We propose a shift from Accumulative Trust to Adaptive, Probabilistic Trust. Instead of asking 'Is this agent good?', our system asks: 'What is the probability that this agent will perform action X successfully in the next transaction, given their full behavioural history, the behaviour of similar agents, and the current state of the network?'"*

2. The Evidence

The need for a trust oracle is not theoretical. It is documented by academic institutions, confirmed by government bodies, quantified by threat intelligence firms, and validated by enterprise practitioners. The full evidence wall with 26 sources is published at agentproof.sh/evidence.

2.1 Academic Research

Agents of Chaos (Stanford, Harvard, MIT, Carnegie Mellon + 6 institutions) -- 38 researchers red-teamed autonomous agents for two weeks in live environments with real email, Discord, file systems, and shell access. Documented 11 failure modes: infrastructure destruction, identity spoofing, social engineering, data exfiltration, partial system takeover. Every failure happened through natural language. No technical exploits required. [arxiv: 2602.20021]

Can LLM Agents Reach Consensus? The Byzantine Problem Revisited -- A single bad actor collapses multi-agent network consensus. In fully benign settings with zero bad actors, LLM agents still fail to converge. The proposed fix -- weighted Byzantine fault tolerance based on agent trustworthiness -- is structurally identical to what AgentProof provides. [arxiv: 2603.01213]

AdapTools -- Adaptive indirect prompt injection with 44-49% attack success rates. We Fixed Jailbreaks, We Did Not Fix Agents -- agents are exploited through normal-looking behaviour in the wrong context, not traditional jailbreaks.

2.2 Government & Threat Intelligence

NIST CAISI (Jan 2026) -- The US Department of Commerce formally solicited industry input on AI agent security. Google Cybersecurity Forecast 2026 identifies 'Shadow Agent Risk' as a priority threat and recommends continuous trust evaluation. CrowdStrike Global Threat Report 2026 reports AI-enabled attacks up 89%. Palisade Research / Anthropic System Cards document models disabling their own shutdown scripts (o3: 79/100 runs), attempting blackmail (Claude Opus 4: 84-96%), and executing insider trades (GPT-4) without instruction.

Craig Riddell, Global Field CISO at Wallarm -- Infostealer malware observed extracting API credentials from live AI agent environments. His conclusion: 'This is not a signature problem. It is a behavioral governance problem. Are we watching what our agents do in motion, or just what credentials they carry?'

2.3 Enterprise & Insurance Validation

Garrett Droege, Willis Towers Watson (National Digital Risk Practice Leader) -- After seeing AgentProof, he outlined the exact actuarial data structure underwriters need to price agent risk tiers and shared a live demo with underwriters. His words: 'The Gold agent rugged me, who pays? question is exactly the coverage gap that will become a headline loss within 24 months.'

Aaron Levie, CEO of Box ($4bn enterprise software) -- Published a full thesis naming agent identity, authentication, and governance as unsolved problems at trillion-agent scale: 'We'll need all new software and platforms to help with these challenges.'

Maragkos Petros, MDX / Avax Team1 -- 'The real open question is whether that reputation layer becomes standard infrastructure for agentic finance, in the same way price oracles became standard infrastructure for DeFi.'

3. ERC-8004 + ERC-8183

ERC-8004 establishes three on-chain registries for AI agents, published by Ava Labs and deployed across 21 EVM-compatible chains via deterministic CREATE2 addresses, plus native Solana program indexing. ERC-8183 (Agentic Commerce Protocol), co-developed by Virtuals Protocol and the Ethereum Foundation dAI team, adds job escrow with a Client-Provider-Evaluator lifecycle. AgentProof composes with both standards via the AgentProofHook -- an IACPHook that gates provider assignment by trust score.

3.1 Identity Registry

ERC-721 NFT-based agent identity. registerAgent(agentURI) mints an identity token with an agent card URI (IPFS, HTTPS, or on-chain data URI). Registration requires a bond as anti-Sybil measure. 55,000+ agents registered across all chains. 58% have broken or invalid URIs -- the registry exists, the integrity layer doesn't. AgentProof is that layer.

3.2 Reputation Registry

On-chain feedback submissions with 1-100 rating scale, feedback URI, task hash, and two structured tags (tag1: assessment dimension, tag2: feedback source). Self-rating prevention enforced at contract level. 24-hour cooldown per reviewer-agent pair.

3.3 Validation Registry

Task validation request/response flow. Requesters submit task hashes, validators respond with validity assessments and proof URIs. Provides the on-chain audit trail for agent task performance.

3.4 Chain Coverage

21 chains indexed: Avalanche, Ethereum, Base, Linea, Polygon, Arbitrum, Optimism, BNB Smart Chain, Scroll, Gnosis, Mantle, Celo, Monad, Abstract, Taiko, MegaETH, SKALE, X Layer, Soneium, Metis, and Solana (native program indexing via ATOM Engine). All EVM chains share deterministic CREATE2 contract addresses. The indexer uses resilient upsert strategies with batch-to-individual fallback and exponential backoff retry logic.

4. Trust Oracle Architecture

The oracle operates as a three-layer system:

Layer 1: Indexing

Multi-chain event indexer scans AgentRegistered, FeedbackSubmitted, ValidationRequested, and ValidationSubmitted events across all 21 chains. Processes in configurable block chunks with catchup mode (500 chunks when far behind vs 50 normal). Persists block pointer for resumability. Resilient upsert: batch -> sub-batch (50) -> individual row fallback with logging.

Layer 2: Evaluation

11-signal composite scoring with Bayesian smoothing, freshness multiplier, 14 automated risk flags, and tier classification. Evaluation results cached in-memory with 300-second TTL. Scoped scores computed per-dimension (tag1) with independent Bayesian smoothing per scope.

Layer 3: Feedback Loop

Oracle submits evaluation results back to the ERC-8004 Reputation Registry as on-chain feedback, creating a verifiable audit trail. Risk levels mapped to on-chain scores: LOW=85, MEDIUM=60, HIGH=30, CRITICAL=10.

5. Scoring Methodology

The composite score (0-100) blends up to 11 weighted signals, Bayesian-smoothed to prevent new agents with a single perfect rating from dominating the leaderboard. Weights dynamically rebalance when optional signals (coding reputation, ERC-8183 job completion) become available.

5.1 Signal Weights

Signal	All	Code	Job	Base	Description
Rating Score	**25%**	27%	28%	30%	Bayesian-smoothed avg (prior=50, k=3)
Validation Score	**11%**	13%	14%	15%	On-chain validation success rate
Account Age	**10%**	11%	11%	12%	Logarithmic maturity curve
Coding Score	**10%**	10%	--	--	GitHub PR merge rate, review quality
Volume Score	**8%**	9%	9%	10%	Logarithmic feedback count
Consistency Score	**8%**	9%	9%	10%	Inverse std dev of ratings
Uptime Score	**8%**	9%	9%	10%	Liveness probe success rate
Job Completion	**8%**	--	8%	--	ERC-8183 job completion rate
Deployer Score	**6%**	7%	7%	8%	Deployer wallet reputation lineage
URI Stability	**6%**	5%	5%	5%	Metadata mutation frequency

All = coding + job active. Code = coding only. Job = job only. Base = neither.

5.2 Additional Tracked Signals

ERC-8183 Job Outcomes -- job completion rate, total jobs as provider, abandonment detection. Indexed from AgentProofHook JobOutcomeRecorded events. Weighted at 8% when available. Delegation Tracking -- success rate, total count, and MTTR (Mean Time To Recovery) when acting as delegate. Reported in evaluations but not yet weighted in composite score. Failure Metrics -- failure count, active unresolved failures, recovery time tracking.

5.3 Bayesian Smoothing

Prior rating: 50.0 (neutral). Smoothing strength k=3. Formula: smoothed = (avg_rating * count + 50 * 3) / (count + 3). An agent with 1 rating of 100 scores 62.5, not 100. An agent with 100 ratings of 100 scores 98.5. This prevents new-agent gaming while allowing established agents to reflect their true performance.

5.4 Freshness Multiplier

All scores are penalised by account age to prevent new-agent sybil attacks: <7 days: 0.70x | 7-30 days: 0.85x | 30-90 days: 0.95x | 90+ days: 1.0x (no penalty). This creates an economic cost to identity rotation -- abandoning a mature identity and starting fresh incurs a 30% score penalty for the first week.

5.5 Tier Thresholds

Diamond	>=85	>=20 feedback	Top-tier verified agents
Platinum	>=72	>=10 feedback	High reliability, established track record
Gold	>=58	>=5 feedback	Good performance, moderate history
Silver	>=42	>=3 feedback	Developing reputation
Bronze	>=30	>=1 feedback	Minimal track record
Unranked	<30	<1 feedback	New or insufficient data

6. Risk Detection & Max Exposure

Every trust evaluation includes a risk assessment with specific flags, a risk level classification (LOW / MEDIUM / HIGH / CRITICAL), and a dollar-denominated max exposure ceiling for insurance underwriting.

6.1 Risk Flags (14)

HIGH_RISK_SCORE	Composite score below safety threshold
CONCENTRATED_FEEDBACK	Majority of feedback from single reviewer
SERIAL_DEPLOYER	Deployer has history of abandoned agents
SUSPICIOUS_VOLATILITY	Score dropped >20 points in monitoring window
LOW_UPTIME	Liveness probe failure rate above threshold
FREQUENT_URI_CHANGES	Agent metadata changed suspiciously often
NEW_IDENTITY	Account age below minimum confidence threshold
LOW_FEEDBACK	Insufficient feedback volume for reliable scoring
UNVERIFIED	No on-chain validations completed
HIGH_FAILURE_RATE	Delegation failure rate above threshold
SLOW_RECOVERY	MTTR exceeds acceptable recovery window
ACTIVE_FAILURE	Unresolved failure event currently active
HIGH_JOB_FAILURE_RATE	ERC-8183 job completion rate <60% with 3+ jobs
JOB_ABANDONMENT	Multiple rejected/expired jobs as provider

6.2 Risk Levels

LOW -- No significant risk flags. Agent has established track record with consistent performance. MEDIUM -- Minor concerns (new identity, low feedback volume). Recommend monitoring. HIGH -- Multiple risk flags active. Recommend caution before delegation. CRITICAL -- Severe risk indicators. Active failures, suspicious volatility, or serial deployer patterns.

6.3 Max Exposure Model

Dollar-denominated trust ceiling calculated from: composite score (base), confidence multiplier (logarithmic feedback volume), age bonus (maturity premium), and validation bonus (verified task history). Provides the actuarial bridge between on-chain reputation and real-world insurance coverage. Underwriters can price agent risk tiers using: transaction volume and velocity, delegation scope, custody relationships, and loss event history with root cause classification.

7. Anti-Identity-Mutation

Identity mutation -- abandoning a tarnished reputation and registering a fresh agent -- is the primary attack vector against any reputation system. AgentProof makes this economically irrational through three mechanisms:

- Freshness Multiplier: New agents suffer a 30% score penalty for 7 days, 15% for 30 days. Starting over costs more than rehabilitating an existing identity.
- Deployer Lineage Tracking: The oracle tracks deployer wallets across all registered agents. A deployer with a history of abandoned or low-scoring agents taints all future registrations (deployer_score signal, 7-8% weight).
- URI Mutation Detection: Frequent changes to agent metadata (name, description, endpoints) are flagged as FREQUENT_URI_CHANGES and penalised through the URI stability signal (5% weight).

8. Sybil Resistance

Preventing gaming, fake reviews, and reputation manipulation through multiple overlapping defenses:

- Bayesian Smoothing (k=3): A single perfect rating scores 62.5, not 100. Gaming requires sustained volume, increasing cost and exposure.
- Feedback Diversity: Concentrated feedback from a single reviewer triggers CONCENTRATED_FEEDBACK risk flag.
- Temporal Consistency: High standard deviation in ratings lowers the consistency score (9-10% weight).
- Deployer Reputation: Sybil deployers who create and abandon agents accumulate negative deployer scores that taint future registrations.
- Anomaly Monitoring: Autonomous oracle job runs every 120 seconds detecting >20-point score drops and flagging SUSPICIOUS_VOLATILITY.
- Registration Bond: ERC-8004 registration requires a bond (0.1 AVAX on Avalanche), making mass identity creation expensive.
- Cross-Chain Correlation: The oracle correlates agents across chains, detecting deployers who spread identities across multiple networks.

9. Protocol Endpoints & Developer Tools

9.1 REST API

Base URL: oracle.agentproof.sh/api/v1

GET /trust/{id}	Full trust evaluation with score breakdown, risk flags, delegation stats
GET /trust/{id}/risk	Focused risk assessment with flag details and risk level
GET /trust/{id}/dimensions	Per-dimension score breakdown
POST /trust/batch	Evaluate up to 500 agents in a single request
GET /agents/trusted	Find trusted agents by category, score, tier, chain
GET /network/stats	Network-wide trust statistics (free, no API key required)

9.2 A2A (Agent-to-Agent)

POST /a2a -- Google Agent-to-Agent protocol. Agents can query trust evaluations using the A2A standard. Agent card published at /.well-known/agent.json.

9.3 MCP Server

POST /mcp -- Model Context Protocol server for Claude, GPT, and other LLM agents. Exposes trust evaluation as an MCP tool that agents can call natively.

9.4 Webhooks

Register webhook URLs to receive real-time notifications when agent scores change. SSRF protection enforced on all registered URLs (private IP blocking, scheme validation, DNS resolution checks).

9.5 TypeScript SDK

@agentproof/sdk v1.1.0 -- Full oracle API support: evaluateTrust(), riskCheck(), dimensions(), batchEvaluate(), findTrusted(), networkStats(). Published on npm.

10. Autonomous Oracle Operations

The oracle runs 8 autonomous background jobs on continuous schedules, maintaining real-time scoring accuracy without manual intervention.

Agent Screening	60s	Screens new agents (batch 200), computes full evaluations, submits on-chain feedback
Anomaly Monitor	120s	Detects >20pt score drops, flags SUSPICIOUS_VOLATILITY, triggers re-evaluation
Liveness Probing	300s	HTTP health checks to agent endpoints (batch 20, 10s timeout per agent)
Failure Metrics	300s	Aggregates failure events, calculates MTTR, identifies active unresolved failures
Network Report	600s	Publishes ecosystem stats via event feed (tier distribution, category averages)
Delegation Sync	600s	Tracks delegation success/failure rates, updates MTTR metrics
Job Outcomes	600s	Syncs ERC-8183 job completion rates, updates job_score signal
GitHub Sync	3600s	Fetches GitHub stats, computes coding_score from PR merge rate and review quality

11. Monetization & API Pricing

All billable endpoints require an API key (X-API-Key header). Keys are registered at oracle.agentproof.sh/integrate. Free endpoints (network stats, key registration) remain unauthenticated. Rate limiting enforced per-key with monthly quotas.

Tier	Price	Limit	Notes
Pay-as-you-go	$0.05/call	1,000 calls/month	No commitment, ideal for testing
Starter	$250/month	10,000 calls/month	Small integrations
Growth	$500/month	50,000 calls/month	Growing protocols
Scale	$1,000/month	200,000 calls/month	High-volume integrations
Enterprise	$2,000/month	Unlimited	Custom SLA, dedicated support

Security: API key middleware validates SHA-256 hashed keys with 60-second cache TTL. SSRF protection on webhooks (private IP blocking, scheme validation, DNS resolution). Rate limiter with selective eviction to prevent memory exhaustion.

12. Competitive Landscape

AgentProof occupies a unique position in the ERC-8004 ecosystem. Existing players fall into two categories, neither of which provides trust evaluation:

Explorers (8004scan, growthepie)

Read-only block explorers that display agent registration data. No scoring, no risk assessment, no API for programmatic trust queries. Analogous to Etherscan -- useful for viewing data, but not for making trust decisions.

Registration SDKs (Agent0)

Developer toolkits for registering agents and submitting feedback on-chain. Essential plumbing, but no intelligence layer. More agents registering through better SDKs increases the need for AgentProof, not reduces it.

AgentProof: The Trust Oracle

The only service that transforms raw on-chain data into actionable trust intelligence. Think Moody's credit ratings, not Etherscan. Protocols pay $0.05/call to query whether an agent is trustworthy before delegating capital or authority. The more agents that register, the more critical trust evaluation becomes.

13. Roadmap

Q1 2026 (Complete)

- 21-chain indexing with 54K+ agents
- 11-signal composite scoring with Bayesian smoothing
- 14 risk flags and 4 risk levels
- ERC-8183 reputation-gated jobs hook (AgentProofHook)
- REST API, A2A, MCP server, webhooks
- TypeScript SDK v1.1.0
- API key gating with 6 pricing tiers
- Autonomous oracle (8 background jobs)
- GitHub coding reputation integration
- Delegation tracking and failure metrics

Q2 2026

- Multi-oracle consensus (2nd oracle deployment)
- Context-aware per-skill trust scores
- Insurance marketplace with staking/claims
- TEE-based validation for high-value evaluations

Q3 2026

- Cross-protocol reputation portability
- Institutional API tier with custom SLAs
- Agent behavioral clustering and regime detection
- Real-time anomaly response (automated downgrades)

Q4 2026

- Full insurance underwriting integration
- Governance token for oracle parameter updates
- Enterprise dashboard for fleet-level agent monitoring
- Expansion to non-EVM chains (Cosmos, Move-based)

14. Conclusion

The agent economy is no longer theoretical. 54,000+ agents are registered on-chain across 21 chains. Circle and Stripe are building payment infrastructure for agent commerce. Enterprise software CEOs are publishing theses on trillion-agent futures. Insurance underwriters are asking how to price agent risk.

What's missing is the trust layer. The ERC-8004 standard provides the registries -- identity, reputation, validation -- but explicitly leaves the integrity layer as an exercise for the ecosystem. AgentProof is that layer.

When a protocol needs to decide whether to delegate capital to an agent, when an insurance underwriter needs to price coverage for an agent fleet, when a multi-agent system needs a deterministic trust anchor instead of another stochastic LLM opinion -- they make a single oracle call.

agentproof.sh | oracle.agentproof.sh/api/v1 | @agentproof/sdk

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